

2021 10%: selected excess 3 at a 185 MeV signal hypothesis

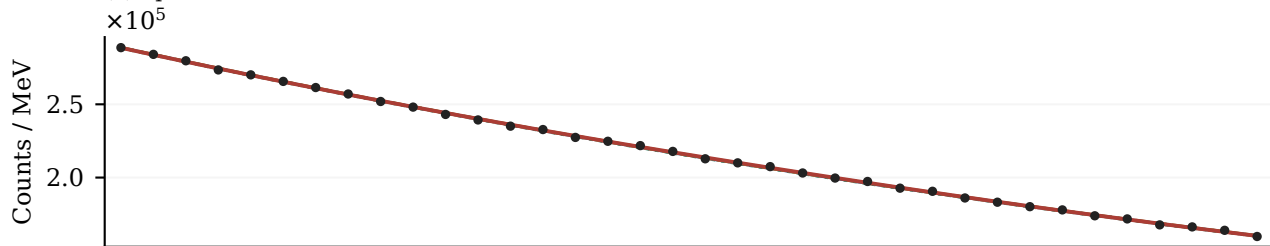
v16 neighboring signal-MC template | $c = 181.340$ MeV, $s = 3.234$ MeV

Signed full-selected yield = $6,063 \pm 3,511$; local asymptotic $Z = 1.73 \sigma$, $p = 0.0421$

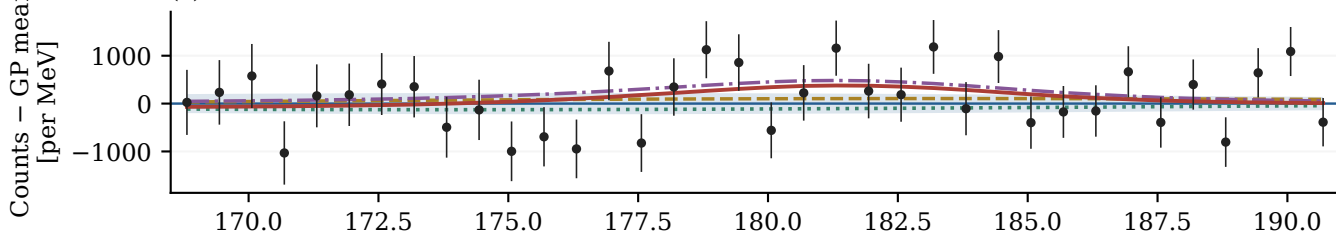
GP background-only toys: local excess rank $p = 0.035$ (34 / 1000 more extreme toys); count 95% interval [0.0237, 0.0472]

- Observed counts
- ⋯ Background in signed S+B fit
- ⬜ Blind region: excluded from GP training
- ⬜ GP mean and ± 1 SD constraint
- Signed signal + background
- Search mass hypothesis
- Background-only fit
- Signed signal (middle panel)
- Signal core center

(a) Spectrum in the fit interval



(b) Residuals relative to the GP mean



(c) Standardized residuals and blind window

